

MARCUS LIANG

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SKILLS SUMMARY

- **AI Models, APIs & Frameworks:** OpenAI API (GPT-5), Claude API, Kimi (Moonshot), Ollama, Nvidia NIM, LangGraph, CrewAI, LangChain
- **Machine Learning & RAG:** PyTorch, Scikit-learn, XGBoost, Random Forest, FAISS (Vector Stores), ClinicalBERT, Model Validation (PSI)
- **Languages & Cloud:** Python, SQL (MySQL), Java, C#, R, JavaScript, Node.js, FastAPI, Docker, Databricks, Apache Spark, AWS, Snowflake
- **BI & Data Analytics:** Pandas, NumPy, Polars, Power BI (DAX), Tableau, Advanced Excel, Vintage Analysis
- **Automation & Workflow:** UiPath, Power Automate, MFlow, GitHub (DevOps)

EXPERIENCE

Business Analyst (Contract), PhillipCapital, Singapore Aug 2026 - Present

- Co-developing an automated trading management system to dynamically adjust buy/sell thresholds based on user credibility profiles.
- Optimizing existing RPA workflows and resolving operational bottlenecks to improve automation uptime across business units.

Business Analyst Intern (Credit Risk), PhillipCapital, Singapore May 2026 - Jul 2026

- Developed an AI-integrated RPA framework to automate standard reporting and parse data to extract actionable business insights.
- Spearheaded integration of mainstream AI tools with UiPath workflows, transforming static reports into dynamic, insight-driven analytics while saving up to 8 man-hours per reporting cycle.
- Gathered and synthesised feedback from key business stakeholders to refine prompt engineering and fine-tuning parameters, improving accuracy and quality of LLM outputs.

Data Scientist Intern, X Star Technology Pte Ltd, Singapore Dec 2024 - Dec 2025

- Engineered predictive repayment models deploying Random Forest and XGBoost to identify loaner's patterns and flag high-risk accounts, improving loan recovery efficiency.
- Developed credit risk ML models using financial and behavioural data, increasing automated approval rates by ~80% in collaboration with cross-functional teams in Beijing.
- Architected interactive Power BI dashboards visualise repayment performance and track KPIs for dealer network, supporting decision-making.
- Executed financial Vintage Analysis grouped by Months on Books (MOB) to evaluate long-term loan performance and forecast future outcomes, such as early settlements.

Cross-Border E-Commerce Intern, Shopee Singapore Pte Ltd, Singapore Sep 2020 - Feb 2021

- Automated key operational processes using Google Sheets and email notifications, reducing manual workload by up to 10 hours per week.
- Developed and implemented SQL queries to extract and analyse critical sales data, empowering data-driven decisions on customer behaviour and sales trends.
- Visualized monthly sales trends on dynamic dashboards to support cross-border commercial strategy.

EDUCATION

National University of Singapore Aug 2023 - Present

School of Computing, Bachelor of Science in Business Analytics (Honours)

- Expected Graduation: Dec 2026
- Specialised in Machine Learning
- Relevant Coursework: BT4221 Advanced Analytics with Big Data Technologies, BT4240 Machine Learning for Predictive Data Analytics, BT3017 Feature Engineering for Machine Learning, CS2040 Data Structures and Algorithms

Ngee Ann Polytechnic Apr 2018 - May 2021

School of InfoComm Technology, Diploma with Merit in Financial Informatics

- Specialised in Financial Analytics
- Relevant Coursework: Deep Learning, Predictive Analytics, Applied Analytics - Finance

KEY PROJECTS

AI Engineer, Agentic Collaborative Reasoning for Cancer Research, National Cancer Centre Singapore Jan 2026 - Apr 2026

- Architected a LangGraph multi-agent system enabling clinicians to query OMOP CDM healthcare data via natural language, running offline via Ollama to assure patient data privacy
- Engineered a Feature Engineering Agent implementing the Refactor → Break Down → Compile decision procedure (ArXiv 2506.09742), improving Kaplan-Meier survival analysis reliability by eliminating column-name guessing
- Designed a Human-in-the-Loop (HITL) workflow with three clinician review checkpoints leveraging LangGraph interrupts, ensuring auditability before any results are delivered.
- Enhanced accuracy of clinical concept mapping by replacing traditional lexical search with a vector-based similarity search engine using ClinicalBERT and FAISS within the OMOP domain.

AI Engineer, Judicial Analytics & Legal NLP, National University Singapore Jan 2026 - Apr 2026

- Structured a custom 3-stage computerised data extraction pipeline utilising the OpenAI GPT-5 API to convert 122 raw, unstructured PDF court judgements into machine-learning-ready JSON records.
- Engineered an "LLM-as-a-Judge" quality assurance auditor, combining a secondary LLM with deterministic regex rules to automatically flag hallucinations and mathematically scrub target leakage.
- Parsed dense judicial narratives into a strict IRAC (Issue, Rule, Application, Conclusion) framework by employing advanced zero-shot and few-shot prompting techniques.
- Developed a custom Hierarchical Attention Network (HAN) for binary classification of corporate liability, utilising Bidirectional GRU layers and SwiGLU activation functions.